

Tutorial 4: MIPS rating and Amdahl's Law

Q1. The MIPS rating of a machine is 300 MIPS, and the CPI is 2. What is the clock rate of the machine?

Solution:

$$\text{MIPS rating} = \text{Clock Rate} / \text{CPI}$$

$$\text{Clock Rate} = 300 \times 10^6 \times 2 = 600 \text{ MHz}$$

Q2. Consider a processor with two instruction classes, A and B, with corresponding CPI values of 1.5 and 2, respectively. The processor runs at a clock rate of 2.5 GHz. For a given program, the instruction counts for the two types of instructions are 50 million and 10 million, respectively. What are the MIPS rating and the CPU time of the processor?

Solution:

$$\begin{aligned}\text{CPI} &= \text{CPU Execution cycles} / \text{Instruction Count} \\ &= (50 \times 1.5 + 10 \times 2) / (50 + 10) \\ &= 95 / 60 = 1.583\end{aligned}$$

$$\begin{aligned}\text{MIPS Rating} &= \text{Clock Rate} / \text{CPI} = (2.5 \times 10^9) / 1.583 = 1.57928 \times 10^9 \\ &= 1579.28 \times 10^6 = 1579.28 \text{ MIPS}\end{aligned}$$

$$\begin{aligned}\text{CPU Time} &= \text{Instruction Count} \times \text{CPI} / \text{Clock Rate} \\ &= (50 + 10) \times 10^6 \times 1.583 / (2.5 \times 10^9) \\ &= 37.992 \times 10^{-3} \text{ sec} \\ &= 37.992 \text{ ms}\end{aligned}$$

Q3. A part of the processor is improved which affects only 25% of the overall tasks. The speed up of the fraction enhanced is 2. Find the overall speedup of the system.

Solution:

Given: Fraction (F) = 25% or 0.25

Fraction Speedup Enhanced (S) = 2

To Find: Overall Speedup

Finding:

$$\text{Overall speedup} = \frac{1}{(1-F) + \frac{F}{S}} = \frac{1}{(1-0.25) + \frac{0.25}{2}} = \frac{1}{0.875} = 1.14$$

Therefore, the overall speed up is 1.14

Q4. A program consists of two loops L1 and L2 respectively. L1 and L2 have 800 and 200 lines of codes respectively. However, the total time of L1 is only 20% and that of L2 is 80%. Calculate the following:-

- Overall speedup when we make L1 90X faster
- Overall speedup when we make L2 10X faster
- Overall speedup when we make L2 50X faster

Solution:

- We make L1 (20% of the execution time of the program) 90X faster,
Here, $F = 0.20$ and $S = 90$

$$\text{Overall speedup} = \frac{I}{(1-F) + \frac{F}{S}} = \frac{I}{(1-0.20) + \frac{0.20}{90}} = \frac{I}{0.8 + 0.0022} = 1.25$$

- We make L2 (80% of the execution time of the program) 10X faster,
Here, $F = 0.80$ and $S = 10$

$$\text{Overall speedup} = \frac{I}{(1-F) + \frac{F}{S}} = \frac{I}{(1-0.80) + \frac{0.80}{10}} = \frac{I}{0.2 + 0.08} = 3.57$$

- We make L2 (80% of the execution time of the program) 50X faster,
Here, $F = 0.80$ and $S = 50$

$$\text{Overall speedup} = \frac{I}{(1-F) + \frac{F}{S}} = \frac{I}{(1-0.80) + \frac{0.80}{50}} = \frac{I}{0.2 + 0.016} = 4.63$$

Q5. A program is running on a processor for which the following instruction mix is observed.

Type	Frequency (%)	CPI
Memory	35	5
Arithmetic	55	2
Control	10	3

Calculate the overall CPI, % of time for each instruction. Suppose we carry the following design enhancements

- Memory is upgraded and the CPI is reduced to 3
- ALU is upgraded and the CPI for arithmetic instruction is reduced to 1
- Branch and Control hardware is upgraded to reduce the CPI to 1

Find the overall performance improvement for each case. Which design enhancement has the most impact on the overall system?

Solution:

Type	F_i	CPI_i	$F_i * CPI_i$	% Time = $\left(\frac{F_i \times CPI_i}{CPI}\right)$
Memory	0.35	5	1.75	0.56
Arithmetic	0.55	2	1.1	0.35
Control	0.10	3	0.3	0.09
Overall CPI			3.15	

- a. Memory is upgraded and the CPI is reduced to 3, Here, $F = 0.56$, and Enhancement factor, $S = 5/3 = 1.67$

$$\text{Overall speedup} = \frac{1}{(1-F) + \frac{F}{S}} = \frac{1}{(1-0.56) + \frac{0.56}{1.67}} = \frac{1}{0.44 + 0.34} = 1.29$$

- b. ALU is upgraded and the CPI for arithmetic instruction is reduced to 1, here, $F = 0.35$, and Enhancement factor, $S = 2/1 = 2$

$$\text{Overall speedup} = \frac{1}{(1-F) + \frac{F}{S}} = \frac{1}{(1-0.35) + \frac{0.35}{2}} = \frac{1}{0.65 + 0.175} = 1.21$$

- c. Branch and Control hardware is upgraded to reduce the CPI to 1, here, $F = 0.09$, and Enhancement factor, $S = 3/1 = 3$

$$\text{Overall speedup} = \frac{1}{(1-F) + \frac{F}{S}} = \frac{1}{(1-0.09) + \frac{0.09}{3}} = \frac{1}{0.91 + 0.03} = 1.06$$

Q6. The execution time of a program on a machine is found to be 100 seconds, out of which 78 seconds is consumed by multiply operations. It is required to make the program run 3 times faster.

- a. By how much must the speed of the multiplier be improved?
- b. Can we make the program run 5 times faster by just improving the multiplier?

Solution:

Given: $F = 78/100 = 0.78$

Overall Speedup = 3

To Find:

Fraction speedup enhanced, $S = ?$

Finding:

a. Overall speedup = $\frac{1}{(1-F) + \frac{F}{S}}$

$$\text{or, } 3 = \frac{I}{(1-0.78) + \frac{0.78}{S}}$$

$$\text{or, } 3 = \frac{I}{\frac{0.22S + 0.78}{S}}$$

$$\text{or, } 3(0.22S + 0.78) = S$$

$$\text{or, } S - 0.66S = 2.34$$

$$\text{or, } S = \frac{2.34}{0.34} = 6.88$$

b. The maximum expected overall speedup is

$$= \frac{I}{(1-F)} = \frac{I}{(1-0.78)} = \frac{I}{0.22} = 4.54$$

Hence, we cannot make the program 5 times faster by just improving the multiplier

Q7. Three processors making companies C1, C2, and C3 release modifications to the three major components (X, Y, and Z) of their processors to improve the execution time. The fraction of the total execution time utilized by a running application on X, Y, and Z are 25%, 55%, and 20% respectively. The speedup gained for the components X, Y, and Z corresponding to the individual companies are outlined below. Calculate the speedup obtained for the processor belonging to each individual company. Rank the companies in terms of the speedup of their processors

	X	Y	Z
C1	1.2	0.3	1.6
C2	0.8	1.5	2.1
C3	1.8	0.6	1.2

Solution:

$$\text{Speedup by company C1} = \frac{I}{(1-0.25-0.55-0.20) + \frac{0.25}{1.2} + \frac{0.55}{0.3} + \frac{0.20}{1.6}} = \frac{I}{0 + 0.20 + 1.83 + 0.125} = \frac{I}{2.155} = 0.46$$

$$\text{Speedup by company C2} = \frac{I}{(1-0.25-0.55-0.20) + \frac{0.25}{0.8} + \frac{0.55}{1.5} + \frac{0.20}{2.1}} = \frac{I}{0 + 0.312 + 0.36 + 0.09} = \frac{I}{0.76} = 1.315$$

$$\text{Speedup by company C3} = \frac{I}{(1-0.25-0.55-0.20) + \frac{0.25}{1.8} + \frac{0.55}{0.6} + \frac{0.20}{1.2}} = \frac{I}{0 + 0.13 + 0.91 + 0.16} = \frac{I}{1.2} = 0.83$$

Therefore speedup in terms of descending order is C2, C3, C1